

UNIT– I VLSI Development Flow overview and System Verilog Basics 6L, 4P

VLSI Technology Overview, IP, Subsystems, and Chips, SOC Architecture, VLSI Design Flow, VLSI Design Verification Flow, Introduction to System Verilog, Data Types and Variables, Operators in System Verilog, Arrays and Associative Arrays, System Verilog Interfaces

UNIT– II Advanced System Verilog Concepts 6L,6P

Procedural Statements and Flow, Control Processes, Tasks, and Functions, Object Oriented Programming concepts, Classes, and Objects, Randomization, Constraints, Inheritance and Polymorphism

UNIT–III System Verilog for Design and Verification 6L,6P

System Verilog for Design: Unique/Priority case, Packages, Hierarchy and Connectivity, synthesizable RTL, SV testbench Architecture, Functional Coverage, Assertions

UNIT– IV EDA tools and Communication Protocols 6L,4P

Setup and Usage of EDA tools relevant (Such as Synopsys VCS/Verdi), APB and SPI Protocol Specifications, Bus Transactions, Use Cases

UNIT– V APB Timer Block Design 6L,10P

APB Specification & Microarchitecture, State Machine and RTL Design, Register and Counter Implementation and Verification using System Verilog (Driver, Monitor, Scoreboard, etc.), with EDA tools, Generate Course Report.

TOTAL: 30L + 30P = 60 PERIODS

COURSEOUTCOMES:

Upon completion of the course, the students will be able to

1. Develop synthesizable RTL designs using System Verilog constructs such as data types, operators, arrays, and interfaces.
2. Apply object-oriented programming concepts in System Verilog to create reusable verification components.
3. Design testbenches using System Verilog constructs, assertions, and coverage metrics for verifying complex digital designs.
4. Use industry-standard EDA tools to simulate, debug, and analyse digital designs and protocols like APB and SPI.
5. Plan, design, and implement a mini project by integrating RTL design and verification concepts using System Verilog and EDA tools.

REFERENCES:

1. Spears, C., & Thornton, G. T., SystemVerilog for Verification: A Guide to Learning the Testbench Language Features, 3rd Edition, Springer, 2012.
2. Bhasker, J., SystemVerilog for Design: A Guide to Using SystemVerilog for Hardware Design and Modeling, 2nd Edition, Springer, 2009.
3. Ben Cohen, SystemVerilog Assertions Handbook, 4th Edition, VhdlCohen Publishing, 2010.
4. Rai, Sudeep Pasricha, On-Chip Communication Architectures: System on Chip Interconnect, 1st Edition, Morgan Kaufmann, 2010.

Lab Experiments:

L1. Design, simulate, and verify SystemVerilog programs using various data types, operators, and procedural constructs.

L2. Design, simulate, and verify SystemVerilog programs using arrays, associative arrays, and interfaces.

L3. Develop, simulate, and verify SystemVerilog programs using tasks, functions, and procedural control statements.

L4. Design reusable verification components using object-oriented programming concepts and interprocess communication

L5. Design and verify constrained-random testbenches using randomization techniques and constraints for functional verification.

L6. Develop and analyze functional coverage models and SystemVerilog assertions to validate digital circuit functionality.

Practical Tasks:

T1. Design, simulate, and verify a synthesizable RTL module using SystemVerilog by applying data types, procedural constructs, arrays, interfaces, tasks, and functions.

T2. Design, simulate, and verify APB and SPI communication protocols using SystemVerilog testbenches, assertions, and functional coverage with industry-standard EDA tools.

T3. Mini Project – Complete RTL Design and Verification using SystemVerilog/ open source tools for advanced digital design and verification

CO	Description of CO	PO	PSO
CO1	Explain the VLSI design and verification flow, SystemVerilog fundamentals, data types, operators, arrays, interfaces, and synthesizable RTL concepts used in digital system design..	----	-----
CO2	Apply advanced SystemVerilog programming concepts including procedural constructs, tasks, functions, object-oriented programming, randomization, constraints, inheritance, and polymorphism to develop reusable verification components.	PO1(3)	PSO1 (3)
CO3	Analyze SystemVerilog constructs to develop synthesizable RTL designs, packages, interfaces, and reusable verification components including testbenches, drivers, and monitors for digital systems.	PO2(3),PO3(2)	PSO1 (3)
CO4	Evaluate digital designs using scoreboards, assertions, functional coverage, hierarchy, connectivity, and verification methodologies to ensure functional correctness and design quality.	PO4(3),PO5(3)	PSO2(3)
CO5	Demonstrate APB and SPI communication protocols using industry-standard EDA tools by performing simulation, debugging, protocol verification, coverage analysis, and interpreting verification results.	PO3(3), PO5(3), PO9(2), PO12(2)	PSO2(3)
CO6	Implement a complete SystemVerilog-based digital system by integrating synthesizable RTL to realize an industry-oriented mini project.	PO5(2), PO9(2), PO12(2)	PSO3(2)

Weightage: Continuous Assessment (CA): 50%, End Semester Examinations: 50%.

Assessment Methodology for CA: Mini Project /Simulation (30%), Practical (30%), Internal Examinations (40%).

Assessment Methodology for Practical: Lab Experiments: 50%, Practical Tasks: 50% (Each student must complete two tasks).

CO \ PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	-	-	-	-	-	-	-	-	-	-	-	-
CO2	3	-	-	-	-	-	-	-	-	-	-	-
CO3	-	3	2	-	-	-	-	-	-	-	-	-
CO4	-	-	-	3	3	-	-	-	-	-	-	-
CO5	-	-	3	-	3	-	-	-	2	-	-	2
CO6	-	-	-	-	2	-	-	-	2	-	-	2
Avg	3	3	2.5	3	2.7	-	-	-	2	-	-	2